

IOQM – 2025 (Date: 28-09-2025)

1. Let ABCD be a quadrilateral in the xy-plane with AB parallel to CD and $AD = BC$. Suppose $A = (0, 0)$, $B = (10, 0)$, $C = (8, 5)$ and $D = (a, b)$. Determine the value of a^2b .
2. A function is defined on the set of positive integers such that if n is odd integer, $f(n) = n - 1$ and if n is an even integers, $f(n) = n^2 - 1$. Determine the sum of all possible values of n such that $f(f(n)) = 99$.
3. Find the number of positive integers n less than or equal to 100 such that n is not divisible by any prime number other than 2 or 3.
4. The six faces of a cubical die are numbered with $2^0, 2^1, 2^2, 2^3, 2^4, 2^5$ in such a way that the product of the numbers on any pair of opposite faces is 2^5 . Two such dice are stacked one on the top of another. If N is the greatest possible sum of the 9 visible numbers (for all such arrangements of dice), find the sum of the squares of the digits of N .
5. Let N be the coefficient of x^{2025} in the expansion of

$$(x + 1)(x^2 + 3)(x^4 + 5)(x^8 + 7) \dots (x^{2024} + 21)$$
 What is the remainder when N is divided by 100?
6. The sum of four distinct prime numbers is 240. If none of the four primes is greater than 70, what is the smallest of the four numbers?
7. How many positive integers $n \leq 100$ are divisible by all positive integers i such that $i^3 \leq n$?
8. Consider 2×3 rectangle made of 6 unit squares. In how many ways can we fill up the six cells using the numbers 1, 2, 3, 4, 5, 6 one each cell, such that any two numbers in adjacent cells (that is, in cells that share a common side) are coprime to each other?
9. Find the largest integer n such that a square of side length n is contained in a circular disc of area 1000.
10. Find the largest positive integer n for which the inequality $\sum_{k=1}^{2n} (-1)^k k^2 < 100$ holds.
11. Let m be a positive integer satisfying the equation:

$$5(2m + 1)(2m + 3)(2m + 5) = \overline{ababab}$$
 Where a and b represent different digits and $\overline{ababab}$ is a six digit number. What is the value of $m + a + b$?
12. Find the number of ordered pairs (m, n) where m and n are positive integers less than or equal to 20000 such that $m^2 + n^4$ is a power of 2.
13. In a convex quadrilateral ABCD, the lengths of the diagonals are 12 and 16 and the line segment joining the midpoints of the opposite sides are of equal length. What is the maximum possible area of the quadrilateral ABCD?

14. The side AB of a square ABCD is 1 and it is also a chord of circle S. The side CD does not intersect S. The length of the tangent CK, drawn from C to S at the point K is 2. If d is the diameter of S, then calculate d^2 .
15. If a, b, c, d are positive integers such that
 $17(abcd + ab + ad + cd + 1) = 20(bcd + b + d)$
 Find $a^2 + b^2 + c^2 + d^2$.
16. If

$$1 - \frac{1}{2 + \frac{1}{3 + \frac{1}{4 + \frac{1}{5 + \frac{1}{6 + \frac{1}{7}}}}}} = \frac{1}{x_1 + \frac{1}{x_2 + \frac{1}{x_3 + \frac{1}{x_4 + \frac{1}{x_5 + \frac{1}{x_6 + \frac{1}{x_7}}}}}}$$

- Where $x_1, x_2, x_3, x_4, x_5, x_6, x_7$ are positive integers, find $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$.
17. There are 100 cards in a box which are numbered 1 to 100. While being blindfolded, Mainak is going to draw one or more cards from the box. After that, he will remove his blindfold and multiply the numbers on these cards. Mainak wants the product of the numbers on the cards drawn to be a multiple of 6. How many cards does he need to draw to make sure that this will happen?
18. In the plane let the positive end of the x-axis is directed towards east and positive end of y-axis be directed towards north. Suppose you are at (0, 0) and wants to (7, 12). At every move you are allowed to move unit length East or unit length towards North from your current position but you are not allowed to visit any point (h, k) where h, k are odd. Find the number of such path n.
19. Find the number of ordered pairs (m, n) where m and n are positive integers such that $1 \leq m, n \leq 50$ and product mn is a perfect square.
20. How many four digit numbers $\overline{abcd}$, with non zero digits a, b, c, d in base 10 are such that $a+c=bd$ and $b+d=ac$?
21. Let $f: R \rightarrow R$ be a function satisfying $4f(3 - x) + 3f(x) = x^2$ for all real x. Find the value of $f(27) - f(25)$ to the nearest integer.
22. Three girls G_1, G_2, G_3 each read four stories S_1, S_2, S_3, S_4 and discuss which one they like. No story is liked by all the three. For each of the three pairs of the girls, there is atleast one story which is liked by the pairs and not liked by the third. Let n be the number of ways in which this is possible. Find the sum of squares of the digits of n.

23. Let P be a point in the interior of a triangle ABC and let AP, BP, CP meets the sides BC, CA and AB in D, E, F respectively. If $\frac{BP}{PE} = \frac{5}{2}, \frac{CP}{PF} = \frac{7}{3}, \text{ and } \frac{AP}{PD} = \frac{p}{q}$, where p, q are natural numbers and $\gcd(p, q)=1$. Find $p + q$.
24. If a and b are positive integers satisfying $4^a + 4a^2 + 4 = b^2$, what is the maximum value of $a + b$?
25. How many natural numbers $n \leq 105$ are such that $7|(2^n - n^2)$?
26. Let ABC be a triangle, D be the midpoint of side BC, O be the circumcenter and H be the orthocenter. If the triangle ODH is equilateral with side length equal to 6 and the area of the triangle ABC can be written as $a\sqrt{b}$, where a and b are positive integers and b is not divisible by the square of any prime, find $a + b$.
27. Consider the collection M of all ordered pairs (a, b) of positive integers a and b satisfy:

$$ab = 406 + 11.lcm(a, b) + 7.gcd(a, b)$$
 What is the smallest possible value of $a + b$?
28. There are 10 members in a delegation. No two of them have the same height. Let N be the number of ways in which they can stand in a line for a photograph such that:
- The leftmost person is the shortest
 - The rightmost person is the tallest, and
 - In the line between the shortest and tallest person, there is exactly one person is shorter than both of his immediate neighbours.
- If N can be written as $100a + b$, where a and b are positive integers less than 100. Find $a + b$.
29. Let ABC be an isosceles triangle with sides 13, 13 and 10. The tangents to the incircle, drawn parallel to the sides intersect the sides in points D, E, F, G, H, I which form a hexagon. If the area of the hexagon DEFGHI is $m + \frac{n}{l}$, where m, n, l are positive integers with $n < l$ and $\gcd(n, l) = 1$. What is $m + n + l$?
30. The properties of a regular dodecagon (a polygon with 12 sides) are coloured either blue or red. Let N be the number of all possible colouring such that no three points of the same colour form the vertices of an equilateral triangle, and no four points of the same colour form the vertices of a square. If N can be written as $N = 100p + q$, where p and q are positive integers less than 100, find $p + q$.

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